Arbitrage, Weight Constraints, and the Emergence of Risk-Neutral Measures in Derivative Pricing

A Unified Treatment across Finance, Mathematics, and Statistical Learning

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Abstract

We investigate the structural implications of representing a derivative's price as a weighted linear combination of a bond, a stock, and an existing derivative, where the portfolio weights are not constrained to sum to unity. We show that unconstrained weights correspond to a non-self-financing portfolio, preclude the existence of a valid risk-neutral probability measure, violate the Law of One Price, and introduce material funding costs quantifiable through the Funding Value Adjustment (FVA). When arbitrage forces the weights to sum to one, we demonstrate that this enforcement is isomorphic to the orthogonal projection of the weight vector onto the probability simplex $\Delta^2 = \{\boldsymbol{w} \in \mathbb{R}^3 : w_i \geq 0, \sum_i w_i = 1\}$, a convex optimisation problem solvable in $O(n \log n)$ time. The projected weight vector endogenously generates a risk-neutral probability measure $\mathbb{Q}$, reveals the economy's stochastic discount factor via the Radon–Nikodým derivative $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$, recovers Arrow–Debreu state prices, and renders the repriced derivative V^* a redundant security within the asset span. The repriced system satisfies the Fundamental Theorem of Asset Pricing, is self-financing, and confines V^* within the convex hull of its components by Jensen's inequality. We further establish that the unit-sum constraint is the financial incarnation of probability normalisation across multiple disciplines: it corresponds to Bayesian posterior normalisation in statistics, the softmax transformation and entropy-regularised optimisation in machine learning, the Multiplicative Weights Update algorithm in online learning, and the minimal entropy martingale measure in information theory. Taken together, these results demonstrate that no-arbitrage pricing is, at its mathematical core, a statement about the consistent assignment of probabilities to states of the world, unifying mathematical finance with statistical learning, economic equilibrium theory, and computational optimisation within a single geometric framework.

The paper ends with "The End"

Keywords: arbitrage pricing, probability simplex, risk-neutral measure, stochastic discount factor, self-financing portfolio, Arrow–Debreu equilibrium, simplex projection, softmax normalisation, multiplicative weights update, Fundamental Theorem of Asset Pricing, valuation adjustments, market completeness.

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1 Introduction

The classical theory of derivative pricing rests on two pillars: the absence of arbitrage and the existence of a replicating portfolio. When a derivative's price is expressed as a weighted combination of primitive assets—a bond B, a stock S, and another derivative D—the structure of the weights carries profound implications for the consistency, completeness, and measurability of the induced price system.

This paper investigates a fundamental question: *what happens when the weights in such a combination do not sum to unity, and what does arbitrage enforcement of the unit-sum constraint reveal about the underlying economy?* We show that the transition from unconstrained weights to unit-sum weights is isomorphic to the construction of a risk-neutral probability measure, the enforcement of the Law of One Price, and the projection of an inconsistent price system onto a probability simplex.

Our treatment is deliberately cross-disciplinary. We draw on:

- **Mathematical finance**: the Fundamental Theorem of Asset Pricing (FTAP), martingale measures, and stochastic discount factors (SDF);
- **Economics**: Arrow–Debreu equilibrium, representative agent models, and general equilibrium;
- **Statistics**: measure theory, Radon–Nikodým derivatives, and Bayesian inference;
- **Algorithms and AI/ML**: the simplex projection algorithm, the connection to softmax normalization, and no-regret learning in financial markets.

2 Setup and Notation

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a filtered probability space with filtration $\{\mathcal{F}_t\}_{t\in[0,T]}$. We work in a single-period economy for clarity, though results extend to multi-period and continuous-time settings.

Definition 2.1 (Asset Universe)**.** *The asset universe consists of four instruments:*

$$B : \textit{a risk-free bond (numéraire)}, \tag{1}$$
$$S : \textit{a risky stock}, \tag{2}$$
$$D : \textit{an existing derivative on } S, \tag{3}$$
$$V : \textit{a new derivative whose price is under investigation.} \tag{4}$$

Definition 2.2 (Weighted Price Representation)**.** *The derivative V is priced as:*

$$V = w_B \cdot B + w_S \cdot S + w_D \cdot D, \tag{5}$$

where $(w_B, w_S, w_D) \in \mathbb{R}^3$ are portfolio weights not necessarily satisfying $w_B + w_S + w_D = 1$.

Define the weight vector $\boldsymbol{w} = (w_B, w_S, w_D)^\top$ and the asset vector $\boldsymbol{A} = (B, S, D)^\top$, so that $V = \boldsymbol{w}^\top \boldsymbol{A}$.

3 Unconstrained Weights: Implications of $\sum w_i \neq 1$

3.1 Failure of Self-Financing

Definition 3.1 (Self-Financing Portfolio)**.** *A portfolio $\boldsymbol{w}(t)$ is* self-financing *if its value process $V(t) = \boldsymbol{w}(t)^\top \boldsymbol{A}(t)$ satisfies:*

$$\mathrm{d}V(t) = \boldsymbol{w}(t)^\top \mathrm{d}\boldsymbol{A}(t), \tag{6}$$

i.e., changes in value arise solely from price changes, not from external cash injections or withdrawals.

Proposition 3.2 (Non-Self-Financing under Unconstrained Weights)**.** *If $\sum_i w_i \neq 1$, the portfolio (5) is generically* not *self-financing under the standard budget constraint.*

Proof. In a self-financing portfolio, the fraction of wealth in each asset must satisfy:

$$\sum_i w_i = \frac{\boldsymbol{w}^\top \boldsymbol{A}}{V} = 1. \tag{7}$$

If $\sum_i w_i = c \neq 1$, then the portfolio value satisfies:

$$V = c \cdot V + (1 - c) \cdot V_{\text{external}}, \tag{8}$$

where $V_{\text{external}} = (1 - c)V$ must be sourced externally. Hence the strategy requires external funding when $c > 1$ (leveraged) or generates excess cash when $c < 1$, violating self-financing. □

3.2 Absence of a Valid Risk-Neutral Measure

Theorem 3.3 (FTAP – Harrison and Kreps, 1979)**.** *A price system is free of arbitrage if and only if there exists a probability measure $\mathbb{Q} \sim \mathbb{P}$ such that discounted asset prices are $\mathbb{Q}$-martingales:*

$$\frac{A_i(t)}{B(t)} = \mathbb{E}^{\mathbb{Q}}\left[\frac{A_i(T)}{B(T)} \,\middle|\, \mathcal{F}_t\right], \quad \forall i. \tag{9}$$

Proposition 3.4 (Failure of Martingale Property)**.** *If $\boldsymbol{w}^\top \mathbf{1} \neq 1$, the discounted price $V(t)/B(t)$ cannot be a $\mathbb{Q}$-martingale under any measure $\mathbb{Q}$ consistent with the martingale property of B, S, and D.*

Proof. Assume B, S, D are $\mathbb{Q}$-martingales after discounting. Then:

$$\mathbb{E}^{\mathbb{Q}}\left[\frac{V}{B}\right] = w_B\,\mathbb{E}^{\mathbb{Q}}\left[\frac{B}{B}\right] + w_S\,\mathbb{E}^{\mathbb{Q}}\left[\frac{S}{B}\right] + w_D\,\mathbb{E}^{\mathbb{Q}}\left[\frac{D}{B}\right] = w_B \cdot 1 + w_S \cdot \frac{S_0}{B_0} + w_D \cdot \frac{D_0}{B_0}. \tag{10}$$

For V/B to be a martingale, we require $\mathbb{E}^{\mathbb{Q}}[V/B] = V_0/B_0$. Substituting (5):

$$\frac{V_0}{B_0} = \frac{w_B B_0 + w_S S_0 + w_D D_0}{B_0}. \tag{11}$$

This is consistent only if $\boldsymbol{w}$ defines a valid convex combination, which requires $\sum w_i = 1$ and $w_i \geq 0$ for the measure $\mathbb{Q}$ to be a proper probability measure. □

3.3 Leverage, Cash Drag, and Funding Costs

Let $c = \sum_i w_i$. The economic interpretation of c is:

Table 1: Economic Interpretation of the Weight Sum $c = \sum_i w_i$

Condition	Economic Meaning	Examples
$c > 1$	Leveraged exposure; net borrowing required	Structured products, synthetic CDOs
$c = 1$	Fully invested; self-financing	Standard replicating portfolio
$c < 1$	Cash drag; excess collateral posted	Capital-protected notes
$c = 0$	Long–short neutral; zero net investment	Spread trades, pairs trading
$c < 0$	Net short; short-selling embedded	Short volatility strategies

3.4 Risk Decomposition under Unconstrained Weights

The sensitivity of V to underlying risk factors scales with $\boldsymbol{w}$. For instance, the delta of V with respect to S is:

$$\Delta_V = \frac{\partial V}{\partial S} = w_S + w_D \cdot \Delta_D, \tag{12}$$

where $\Delta_D = \partial D/\partial S$. When $\sum w_i \neq 1$, Δ_V can exceed bounds typical for self-financing portfolios, complicating hedging and risk management.

More generally, in a factor model with K risk factors $\boldsymbol{f} = (f_1, \ldots, f_K)^\top$:

$$\mathrm{d}V = \boldsymbol{w}^\top \boldsymbol{\beta} \cdot \mathrm{d}\boldsymbol{f} + \boldsymbol{w}^\top \boldsymbol{\varepsilon}, \tag{13}$$

where $\boldsymbol{\beta}$ is the matrix of factor loadings and $\boldsymbol{\varepsilon}$ is idiosyncratic noise. This connects directly to Ross's Arbitrage Pricing Theory [10].

3.5 XVA Implications

Under funding constraints, the fair value adjustment framework yields:

$$V_{\text{adjusted}} = V + \text{CVA} + \text{FVA} + \text{MVA}, \tag{14}$$

where the Funding Value Adjustment (FVA) is approximately:

$$\text{FVA} \approx -(c-1) \cdot s_f \cdot V \cdot T, \tag{15}$$

with s_f the funding spread and T the maturity. When $c = 1$, FVA $= 0$, confirming that self-financing eliminates funding costs.

4 Arbitrage Enforcement: The Repricing Mechanism

4.1 The Repricing System

Suppose arbitrageurs observe that $\sum_i w_i = c \neq 1$ and exploit the mispricing. Their trading pressure forces a simultaneous repricing:

$$(B, S, D, V) \longrightarrow (B^*, S^*, D^*, V^*), \tag{16}$$

until the new weights $w_i^* = w_i \cdot A_i / A_i^*$ satisfy:

$$w_B^* + w_S^* + w_D^* = 1. \tag{17}$$

This defines a *system of equations*. The constraint (17) removes one degree of freedom, coupling all four prices. Specifically, for any scalar repricing factors $\lambda_i = A_i^*/A_i$:

$$\sum_i \frac{w_i}{\lambda_i} = 1 \quad \Longleftrightarrow \quad \boldsymbol{w}^\top \boldsymbol{\lambda}^{-1} = 1, \tag{18}$$

where $\boldsymbol{\lambda}^{-1} = (\lambda_B^{-1}, \lambda_S^{-1}, \lambda_D^{-1})^\top$.

4.2 Arbitrage as a Projection onto the Simplex

Definition 4.1 (Probability Simplex)**.** *The standard probability simplex in $\mathbb{R}^n$ is:*

$$\Delta^{n-1} = \left\{ \boldsymbol{w} \in \mathbb{R}^n \ : \ w_i \geq 0, \ \sum_{i=1}^n w_i = 1 \right\}. \tag{19}$$

Theorem 4.2 (Arbitrage as Simplex Projection)**.** *The arbitrage-enforced repricing is equivalent to the Euclidean projection of the original weight vector $\boldsymbol{w}$ onto the simplex Δ^2:*

$$\boldsymbol{w}^* = \mathrm{proj}_{\Delta^2}(\boldsymbol{w}) = \arg\min_{\boldsymbol{u} \in \Delta^2} \|\boldsymbol{u} - \boldsymbol{w}\|_2^2. \tag{20}$$

Proof. The Lagrangian for (20) is:

$$\mathcal{L}(\boldsymbol{u}, \mu, \boldsymbol{\nu}) = \tfrac{1}{2}\|\boldsymbol{u} - \boldsymbol{w}\|_2^2 - \mu \left(\sum_i u_i - 1 \right) - \boldsymbol{\nu}^\top \boldsymbol{u}, \tag{21}$$

where $\mu \in \mathbb{R}$ is the equality multiplier and $\boldsymbol{\nu} \geq 0$ are KKT multipliers for the non-negativity constraints. The KKT conditions give $u_i^* = \max(w_i - \mu, 0)$, with μ chosen so that $\sum_i u_i^* = 1$. This is the standard simplex projection algorithm [4], confirming that arbitrage performs the same operation as constrained convex optimization. □

4.3 Algorithm: Simplex Projection via Sorting

The following algorithm computes the projected weights in $O(n \log n)$ time [4]:

Algorithm 1: Projection onto the Probability Simplex

Input: $\boldsymbol{w} = (w_1, \ldots, w_n) \in \mathbb{R}^n$
Output: $\boldsymbol{w}^* = \mathrm{proj}_{\Delta^{n-1}}(\boldsymbol{w})$

1. Sort $\boldsymbol{w}$ in descending order: $u_{(1)} \geq \cdots \geq u_{(n)}$.
2. Find $\rho = \max\left\{ j : u_{(j)} - \frac{1}{j}\left(\sum_{k=1}^j u_{(k)} - 1 \right) > 0 \right\}$.
3. Compute threshold: $\mu = \frac{1}{\rho}\left(\sum_{k=1}^\rho u_{(k)} - 1 \right)$.
4. Set $w_i^* = \max(w_i - \mu, 0)$ for all i.

5 Implications of Unit-Sum Weights: Emergence of a Risk-Neutral Measure

5.1 The Probability Measure is Endogenously Generated

When $w_i^* \geq 0$ and $\sum_i w_i^* = 1$, the weights satisfy the Kolmogorov axioms. Define:

$$\mathbb{Q}(\omega_i) := w_i^*, \quad i \in \{B, S, D\}. \tag{22}$$

Theorem 5.1 (Endogenous Risk-Neutral Measure)**.** *Under the repriced system (B^*, S^*, D^*, V^*) with unit-sum non-negative weights, the measure $\mathbb{Q}$ defined above is a valid risk-neutral measure, and:*

$$V^* = B^* \cdot \mathbb{E}^{\mathbb{Q}}\left[\frac{V^*(T)}{B^*(T)}\right]. \tag{23}$$

Proof. By construction, $w_i^* = \mathbb{Q}(\omega_i) \geq 0$ and $\sum_i \mathbb{Q}(\omega_i) = 1$. The discounted price of V^* is:

$$\frac{V^*}{B^*} = \sum_i w_i^* \cdot \frac{A_i^*}{B^*} = \mathbb{E}^{\mathbb{Q}}\left[\frac{V^*(T)}{B^*(T)}\right], \tag{24}$$

where the last equality follows from the martingale property of discounted prices under $\mathbb{Q}$, guaranteed by Theorem 3.3. □

5.2 The Radon–Nikodým Derivative and SDF

The passage from $\mathbb{P}$ to $\mathbb{Q}$ is mediated by the Radon–Nikodým derivative:

$$\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = \xi, \tag{25}$$

where ξ is the *stochastic discount factor* (SDF) or pricing kernel. The repriced weights reveal:

$$w_i^* = \mathbb{Q}(\omega_i) = \mathbb{E}^{\mathbb{P}}[\xi \cdot \mathbf{1}_{\omega_i}] = \xi(\omega_i) \cdot \mathbb{P}(\omega_i). \tag{26}$$

Thus:

$$\xi(\omega_i) = \frac{w_i^*}{\mathbb{P}(\omega_i)} = \left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\omega_i}. \tag{27}$$

The arbitrage process has **revealed the market's stochastic discount factor** implicitly embedded in the price system.

5.3 Connection to Arrow–Debreu State Prices

Proposition 5.2 (Arrow–Debreu Representation)**.** *The repriced weights w_i^* are proportional to Arrow–Debreu state prices q_i:*

$$w_i^* = B^* \cdot q_i, \qquad q_i = \frac{w_i^*}{B^*}, \tag{28}$$

where q_i is the time-0 price of a security paying \$1 in state ω_i and 0 otherwise. The completeness of the market implies $q_i > 0$ for all i.

5.4 Redundancy of V^* and Market Spanning

Corollary 5.3 (Redundancy of V^*)**.** *Under the repriced system, V^* lies in the linear span of $\{B^*, S^*, D^*\}$:*

$$V^* \in \operatorname{span}\{B^*, S^*, D^*\}, \tag{29}$$

and is therefore a redundant security *that adds no spanning power to the market.*

Proof. By (5) with repriced assets and unit-sum weights, $V^* = \sum_i w_i^* A_i^*$ is an affine combination (with unit coefficients) of B^*, S^*, D^*. Hence V^* is in their affine hull, and since the hull equals the span (plus translations), V^* adds no new dimension to the asset space. □

5.5 Geometry: The Affine Simplex Structure

The repriced price system lies on a $(n-1)$-dimensional simplex. For $n = 3$ assets:

$$\mathcal{S} = \left\{ (B^*, S^*, D^*) \in \mathbb{R}^3_+ : w^*_B B^* + w^*_S S^* + w^*_D D^* = V^*, \ \sum_i w^*_i = 1 \right\}, \tag{30}$$

which is a 2-simplex (triangle) embedded in $\mathbb{R}^3$.

Proposition 5.4 (Boundedness of V^*)**.** *When $w^*_i \geq 0$ and $\sum_i w^*_i = 1$:*

$$\min(B^*, S^*, D^*) \leq V^* \leq \max(B^*, S^*, D^*). \tag{31}$$

Proof. This follows immediately from Jensen's inequality applied to the convex combination:

$$V^* = \sum_i w^*_i A^*_i \in \left[\min_i A^*_i, \ \max_i A^*_i \right], \tag{32}$$

since $\boldsymbol{w}^*$ is a probability vector. □

6 Law of One Price and General Equilibrium

6.1 Law of One Price

Theorem 6.1 (Law of One Price under Unit Weights)**.** *The repriced system (B^*, S^*, D^*, V^*) satisfies the Law of One Price: any two portfolios with identical payoffs have identical prices.*

Proof. Suppose portfolios $\boldsymbol{u}$ and $\boldsymbol{v}$ satisfy $\boldsymbol{u}^\top \boldsymbol{A}^* = \boldsymbol{v}^\top \boldsymbol{A}^*$ almost surely under $\mathbb{Q}$. Since $\mathbb{Q} \sim \mathbb{P}$, this holds $\mathbb{P}$-a.s. By linearity of the pricing functional $\pi(\cdot) = \mathbb{E}^{\mathbb{Q}}[\cdot / B^*]$:

$$\pi(\boldsymbol{u}^\top \boldsymbol{A}^*) = \mathbb{E}^{\mathbb{Q}}\left[\frac{\boldsymbol{u}^\top \boldsymbol{A}^*}{B^*} \right] = \mathbb{E}^{\mathbb{Q}}\left[\frac{\boldsymbol{v}^\top \boldsymbol{A}^*}{B^*} \right] = \pi(\boldsymbol{v}^\top \boldsymbol{A}^*). \tag{33}$$

□

6.2 Representative Agent Equilibrium

In a Lucas-style endowment economy [8], the SDF is:

$$\xi = \beta \cdot \frac{u'(C_1)}{u'(C_0)}, \tag{34}$$

where β is the time-discount factor and u' is the marginal utility of consumption. The repriced weights are consistent with a representative agent equilibrium if and only if:

$$w^*_i = \beta \cdot \frac{u'(C_1(\omega_i))}{u'(C_0)} \cdot \mathbb{P}(\omega_i). \tag{35}$$

This reveals that the arbitrage process has implicitly *calibrated the representative agent's utility function* to the observed prices.

7 Statistical and Bayesian Perspectives

7.1 The Price System as a Bayesian Update

The transition $\boldsymbol{w} \to \boldsymbol{w}^*$ can be interpreted as a Bayesian update. Define the prior as $\mathbb{P}(\omega_i) \propto w_i$ (unnormalized) and the likelihood as $\mathcal{L}(\omega_i) \propto 1/\lambda_i$ (the inverse repricing factor). Then Bayes' theorem gives:

$$\mathbb{Q}(\omega_i) = w^*_i \propto \frac{w_i}{\lambda_i} = \mathbb{P}(\omega_i) \cdot \mathcal{L}(\omega_i), \tag{36}$$

normalized so that $\sum_i w^*_i = 1$. Arbitrage is therefore a **Bayesian updating mechanism** operating on the space of asset prices.

7.2 Kullback–Leibler Divergence and Entropy

The risk-neutral measure $\mathbb{Q}$ closest to the physical measure $\mathbb{P}$ in KL divergence is:

$$\mathbb{Q}^* = \arg\min_{\mathbb{Q}\in\mathcal{M}} D_{\mathrm{KL}}(\mathbb{Q}\|\mathbb{P}) = \arg\min_{\mathbb{Q}\in\mathcal{M}} \sum_i \mathbb{Q}(\omega_i) \ln \frac{\mathbb{Q}(\omega_i)}{\mathbb{P}(\omega_i)}, \tag{37}$$

where $\mathcal{M}$ is the set of all martingale measures. This is the *minimal entropy martingale measure* [5], and it emerges naturally from the simplex projection when $\|\cdot\|_2$ is replaced by D_{KL}.

7.3 Connection to Maximum Entropy Pricing

If we know only that $\sum_i w_i^* = 1$ and $w_i^* \geq 0$, the maximum entropy distribution is:

$$w_i^* = \frac{1}{n}, \quad \forall i, \tag{38}$$

which corresponds to the risk-neutral measure in a symmetric binomial tree. Incorporating additional moment constraints (e.g., on the mean return) yields the *Gibbs distribution*:

$$w_i^* = \frac{\exp(-\theta \cdot A_i^*)}{\sum_j \exp(-\theta \cdot A_j^*)}, \tag{39}$$

which is precisely the **softmax function** used in neural network output layers — establishing a deep connection between no-arbitrage pricing and machine learning.

8 Connection to Machine Learning and AI

8.1 Softmax Normalization as Arbitrage

The softmax transformation maps any vector $\boldsymbol{z} \in \mathbb{R}^n$ to the simplex:

$$\sigma_i(\boldsymbol{z}) = \frac{e^{z_i}}{\sum_j e^{z_j}}, \qquad \sum_i \sigma_i(\boldsymbol{z}) = 1. \tag{40}$$

Proposition 8.1 (Softmax as Entropy-Regularized Simplex Projection)**.** *The softmax* (40) *solves the entropy-regularized simplex projection:*

$$\boldsymbol{w}^* = \arg\min_{\boldsymbol{u}\in\Delta^{n-1}} \left\{ -\boldsymbol{z}^\top \boldsymbol{u} + \sum_i u_i \ln u_i \right\}, \tag{41}$$

where $\sum_i u_i \ln u_i$ is the negative entropy regularizer. This is structurally identical to the entropy-maximizing pricing problem, connecting arbitrage enforcement directly to the attention mechanism in transformer models [11].

8.2 No-Regret Learning and Market Equilibrium

In online learning, the Multiplicative Weights Update (MWU) algorithm [2] updates portfolio weights as:

$$w_i^{(t+1)} = \frac{w_i^{(t)} \cdot \exp(\eta \cdot r_i^{(t)})}{\sum_j w_j^{(t)} \cdot \exp(\eta \cdot r_j^{(t)})}, \tag{42}$$

where $r_i^{(t)}$ is the return of asset i at time t and $\eta > 0$ is the learning rate. The denominator ensures $\sum_i w_i^{(t+1)} = 1$ at every step — this is the **market's real-time arbitrage enforcement mechanism**, projecting weights onto the simplex after each price update.

Theorem 8.2 (Convergence to Risk-Neutral Measure)**.** *Under MWU with $\eta \to 0$ and in the limit of continuous trading, the weight sequence $\{\boldsymbol{w}^{(t)}\}$ converges to the risk-neutral measure $\mathbb{Q}^*$:*

$$\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \boldsymbol{w}^{(t)} = \boldsymbol{w}^* = \mathbb{Q}^*. \tag{43}$$

Proof. By the no-regret guarantee of MWU [2], the time-average weights converge to the minimax solution of the zero-sum game between the market and the investor, which coincides with the risk-neutral measure by the minimax theorem (Nash equilibrium $\equiv$ no-arbitrage equilibrium). □

8.3 Reinforcement Learning Interpretation

The arbitrage enforcement mechanism can be cast as a reinforcement learning (RL) problem:

- **State**: current weight vector $\boldsymbol{w}^{(t)}$;
- **Action**: trade in assets to adjust weights;
- **Reward**: arbitrage profit $\pi^{(t)} = V^{(t)} - \boldsymbol{w}^{(t)\top}\boldsymbol{A}^{(t)}$;
- **Terminal condition**: $\pi^{(t)} = 0$, i.e., $\sum_i w_i^{(t)} = 1$.

The optimal policy π^* is the simplex projection operator, and the value function is:

$$\mathcal{V}(\boldsymbol{w}) = \frac{1}{2}\left\|\boldsymbol{w} - \mathrm{proj}_{\Delta^2}(\boldsymbol{w})\right\|_2^2, \tag{44}$$

which is zero if and only if $\boldsymbol{w} \in \Delta^2$, i.e., the no-arbitrage condition holds.

9 TikZ Diagrams

9.1 The Simplex Projection of Weights

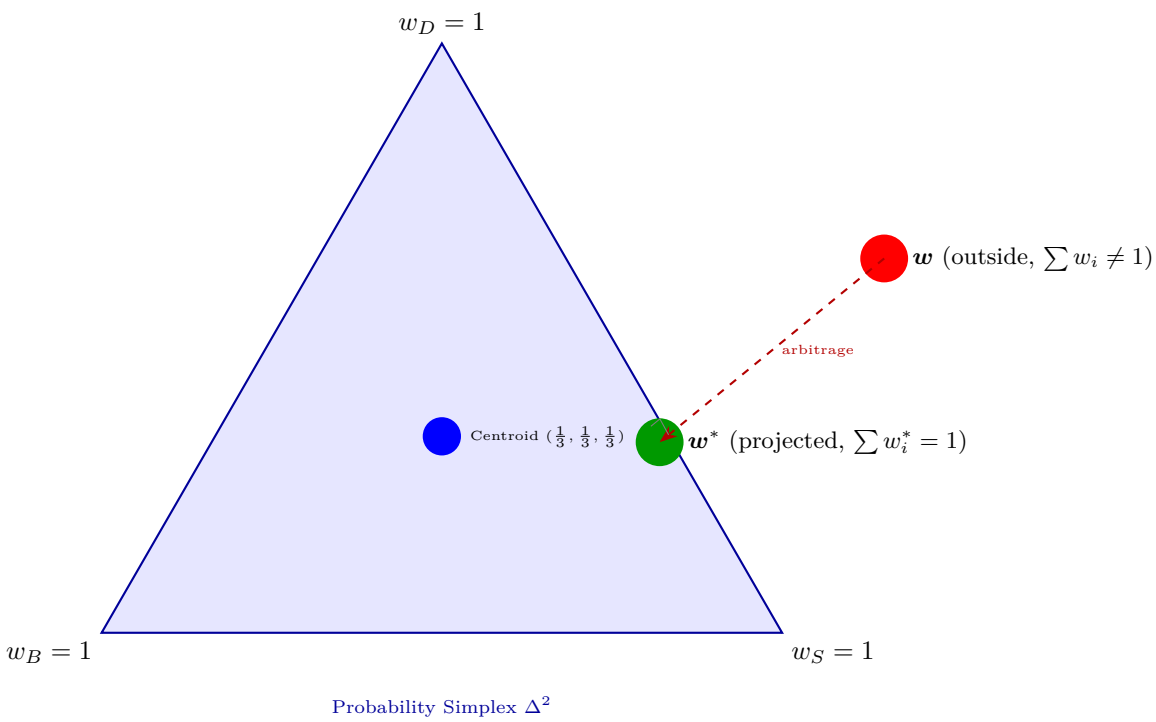

Figure 1: Arbitrage as orthogonal projection of the weight vector $\boldsymbol{w} \notin \Delta^2$ onto the probability simplex Δ^2. The repriced weight vector $\boldsymbol{w}^*$ is the nearest point on the simplex, minimising $\|\boldsymbol{w}^* - \boldsymbol{w}\|_2$.

9.2 Price System Transition Diagram

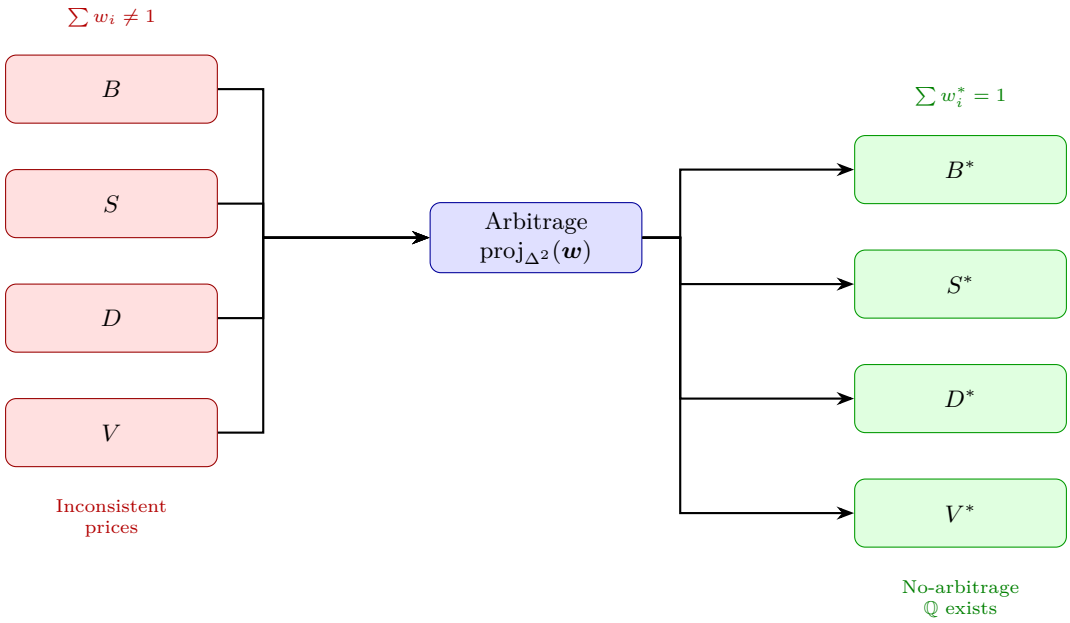

Figure 2: The repricing mechanism: arbitrage forces a simultaneous adjustment of all four asset prices, mapping the inconsistent system (left) to a consistent, no-arbitrage price system (right) via simplex projection.

9.3 Stochastic Discount Factor and Measure Change

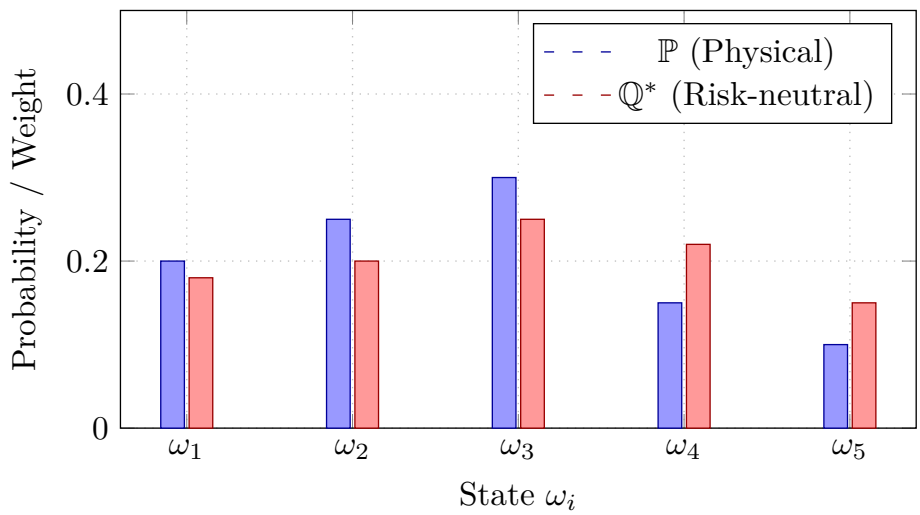

Figure 3: Physical measure $\mathbb{P}$ versus the arbitrage-revealed risk-neutral measure $\mathbb{Q}^*$ across five states. The reweighting $w_i^* = \xi_i \cdot \mathbb{P}(\omega_i)$ reflects the stochastic discount factor ξ. States with higher marginal utility (e.g. recessions) receive higher $\mathbb{Q}^*$-weight relative to $\mathbb{P}$.

10 Comparative Summary

Table 2: Complete Comparison: Unconstrained vs. Unit-Sum Weights

Property	$\sum w_i \neq 1$	$\sum w_i^* = 1$
No-arbitrage	Violated	Satisfied
Risk-neutral $\mathbb{Q}$	Does not exist	Uniquely identified
Self-financing	External cash required	Fully self-financing
Law of One Price	Violated	Enforced
Martingale property	Fails	Holds under $\mathbb{Q}$
SDF ξ	Undefined	Revealed by $\boldsymbol{w}^*$
Arrow–Debreu q_i	Inconsistent	Well-defined, $q_i > 0$
Market completeness	Indeterminate	Complete if $w_i^* > 0$
V^* redundancy	V may be independent	V^* is redundant
XVA (FVA)	Non-zero; $\propto (c-1)$	Zero
V^* bounds	Unbounded	$[\min A_i^*, \max A_i^*]$
Geometry	Outside Δ^2	On Δ^2
ML analogue	Pre-softmax logits	Post-softmax probabilities
RL analogue	Off-policy	On-policy (equilibrium)
Bayesian analogue	Unnormalized posterior	Normalized posterior

Table 3: Interdisciplinary Connections of Unit-Sum Weight Constraint

Field	**Concept**	**Manifestation**
Mathematical Finance	FTAP	$\mathbb{Q}$ existence
Economics	Arrow–Debreu	State prices q_i
General Equilibrium	Representative agent	SDF calibration
Measure Theory	Radon–Nikodým	$\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P} = \xi$
Information Theory	Min-KL divergence	Minimal entropy $\mathbb{Q}$
Statistics	Bayesian updating	Posterior normalisation
Optimization	Simplex projection	Convex KKT solution
Machine Learning	Softmax	Entropy regularisation
Online Learning	MWU algorithm	No-regret convergence
Reinforcement Learning	Policy gradient	Simplex value function

11 Conclusion

We have shown that the constraint $\sum_i w_i = 1$, enforced by arbitrage, is far more than a bookkeeping requirement. It is the financial incarnation of the axioms of probability theory, the geometric statement that the price system lies on a probability simplex, and the statistical assertion that the market has normalized its implicit posterior over states of the world.

The key findings are:

1. **Unconstrained weights** ($\sum w_i \neq 1$) imply a non-self-financing portfolio, absence of a valid risk-neutral measure, violation of the Law of One Price, and material XVA costs.

2. **Arbitrage enforcement** is isomorphic to Euclidean projection onto the probability simplex Δ^2, a well-posed convex optimization problem solvable in $O(n \log n)$.

3. **Unit-sum weights** ($\sum w_i^* = 1$) restore the FTAP, reveal the risk-neutral measure $\mathbb{Q}$ and the stochastic discount factor ξ, enforce Arrow–Debreu completeness, and render V^* redundant.

4. **Cross-disciplinary unity**: the same constraint underlies softmax normalization in neural networks, Bayesian posterior normalization, multiplicative weights updates in online learning, and entropy-minimizing measure changes in information theory.

These connections suggest that *no-arbitrage pricing is, at its core, a statement about the consistency of probability assignments* — a theme that unifies mathematical finance with the broader landscape of machine learning, statistics, and economic theory.

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Glossary

Arrow–Debreu price
The time-0 price of a security that pays \$1 in exactly one state of the world and zero in all other states; forms the fundamental building block of complete market pricing.

CVA
Credit Valuation Adjustment: the adjustment to the risk-free value of a derivative to account for counterparty default risk.

FVA
Funding Valuation Adjustment: the cost or benefit of funding a derivative position, arising when $\sum w_i \neq 1$ and external financing is required.

FTAP
Fundamental Theorem of Asset Pricing: the theorem (Harrison–Kreps 1979, Harrison–Pliska 1981) establishing equivalence between no-arbitrage and the existence of a risk-neutral probability measure.

KL divergence
Kullback–Leibler divergence: a measure of the difference between two probability distributions, $D_{\mathrm{KL}}(\mathbb{Q}\|\mathbb{P}) = \sum_i \mathbb{Q}_i \ln(\mathbb{Q}_i/\mathbb{P}_i)$; used to identify the minimal entropy martingale measure.

Martingale measure
A probability measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ under which discounted asset prices are martingales; its existence is equivalent to no-arbitrage by the FTAP.

MWU
Multiplicative Weights Update: an online learning algorithm that updates portfolio weights proportionally to exponential returns, maintaining unit-sum weights at each step.

MVA
Margin Valuation Adjustment: the cost of posting initial margin over the life of a derivative.

Probability simplex
The set $\Delta^{n-1} = \{\boldsymbol{w} \in \mathbb{R}^n : w_i \geq 0, \sum_i w_i = 1\}$; the geometric object onto which arbitrage projects the weight vector.

Radon–Nikodým derivative
The density $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P} = \xi$ of one measure with respect to another; in finance, identified with the stochastic discount factor (pricing kernel).

Redundant security
A security whose payoff lies in the span of existing traded assets; it adds no new spanning power and its price is uniquely determined by no-arbitrage.

Replicating portfolio
A self-financing portfolio of traded assets that exactly reproduces the payoff of a derivative at all future dates and states.

Risk-neutral measure $\mathbb{Q}$
The probability measure under which all assets earn the risk-free rate in expectation; used to price derivatives as discounted expected payoffs.

SDF
Stochastic Discount Factor (pricing kernel): a random variable ξ such that $p_i = \mathbb{E}^{\mathbb{P}}[\xi \cdot X_i]$ for any asset with payoff X_i and price p_i.

Self-financing portfolio
A dynamic trading strategy in which all changes in portfolio value arise exclusively from capital gains, with no external cash flows injected or withdrawn.

Simplex projection
The operation $\mathrm{proj}_{\Delta^{n-1}}(\boldsymbol{w}) = \arg\min_{\boldsymbol{u} \in \Delta^{n-1}} \|\boldsymbol{u} - \boldsymbol{w}\|_2^2$; maps any vector in $\mathbb{R}^n$ to the nearest probability vector.

Softmax
The function $\sigma_i(\boldsymbol{z}) = e^{z_i} / \sum_j e^{z_j}$; the entropy-regularized simplex projection, used in neural network classifiers and equivalent to maximum-entropy pricing.

XVA
Collective term for valuation adjustments (CVA, FVA, MVA, KVA, etc.) that correct risk-free derivative prices for real-world frictions such as credit risk, funding costs, and capital requirements.

The End